# Pathology

## High-Yield Clinical & Systemic Pathology Compendium

Comprehensive Diagnostic Tables, Histological Hallmarks & Clinical Pearls

## PM2493. Neurocutaneous syndomes

Von Hippel Lindau Syndrome:\n

- Autosomal dominant
- Gene defect: VHL gene, tumor suppressor gene on chromosome 3
- Conditions associated with it include: <

## PM2482. Histological Features of CNS Tumors

Pilocytic Astrocytomas:\n

- Biphasic architecture - loose microcystic component and dense fibrillary areas
- Bipolar cells (tumor cells): thin, long hairlike processes that are **GF**

## PM2430. FAB classification of ALL

## PM2357. Types of calcifications

## PM2356. Intracellular accumulations

## PM2355. Types of granulomas

## PM2353. Types of giant cells

## PM2321. Classification and WHO Grading of Brain tumors

## PM2316. Synucleinopathies and tauopathies

- **Parkinson's disease**
- **Dementia with Lewy bodies**
- **Multiple system atrophy**
- **REM behav**

## PM2315. CNS infections and regions commonly affected

## PM2279. Seminomatous germ-cell tumors (GCTs) vs Non-seminomatous GCT

**AFP: Alpha-fetoprotein\nHCG: Human Chorionic Gonadotropin**

## PM2274. WHO 2016 Classification of Renal Cell Tumors

- **Clear-cell renal cell carcinoma**
- **Multilocular cystic renal neoplasm of low malignant potential**
- **Papillary renal cell carcinoma**
- **Hereditary leiomyomatosis and renal cel**

## PM2270. Inherited causes of nephrotic syndrome

**FSGS: Focal segmental glomerulosclerosis\nNOTE: Inherited forms of nephrotic syndrome are invariably resistant to immunosuppressants (steroid resistant)**

## PM2265. Onion Skin Appearance in Pathology

**Onion skin appearance:\nA pattern characterised by concentric laminations of differing radiologic or histologic densities. It is seen in:\n**

1. **Primary sclerosing cholang**

## PM2241. Microvesicular Steatosis vs. Macrovesicular Steatosis

## PM2227. Necrosis vs Apoptosis

## PM2209. The most commons in GIT

## PM2184. Precursor Lesions of Hepatocellular Carcinoma

**NAFLD: Non-alcoholic fatty liver disease\nHBV: Hepatitis B virus**

## PM2174. Human Papilloma Virus (HPV) and Associated Lesions

**LSIL: Low-grade squamous intraepithelial lesion\nHSIL: High-grade squamous intraepithelial lesion**

## PM2173. Pattern of nuclear body suggests the type of antibody present in patient's serum.

**Five basic patterns of nuclear fluorescence is recognized for anti-nuclear antibodies:**

## PM2112. Tumor suppressor genes

## PM2067. WHO Classification of Tumors of the Urothelial Tract (2016)

**1) Urothelial (transitional) tumors (most common)\n**

- **Invasive**
- **Non-invasive**
  - **Carcinoma in situ**
  - **Urothelial papilloma**
  - **Inverted urothelial papilloma**
  - **Papi**

## PM2061. Role of CK7 and CK20 Markers in Tumors of Unknown Primary

**Cytokeratin is an epithelial marker. A combination of CK7 and CK20 is used to identify the organ of origin.**

## PM2020. Location of Important Genes on Chromosome

## PM1993. Mechanisms of Immune Evasion by Tumors

**Tumors evade host immunity by the following mechanisms:\n**

1. **Selective outgrowth of tumor cells that do not express tumor antigens**
2. **Loss or reduced expression**

## PM1992. Tumor antigens recognized by CD8+ T cells

**Antitumor activity is mediated predominantly by cell-mediated mechanisms. Tumor antigens are presented on the cell surface by MHC class I molecules and are recognized by CD8+ CTLs.**

## PM1986. WHO classification of Lymphoid Neoplasms

## PM1970. Various actions of Principal Mediators of Inflammation

**Note: Thromboxane A2 (TXA2) is a prostaglandin, which is a potent vasoconstrictor**

## PM1969. Principal Mediators of Inflammation and Their Source

## PM1965. International Staging System (ISS) for multiple myeloma:

**International Staging System (ISS) for multiple myeloma: ISS is the most widely used method of assesing prognosis in multiple myeloma.**

## PM1896. Mallory Bodies vs. Councilman Bodies

**Mallory bodies seen as clumped, amorphous eosinophilic material within ballooned hepatocytes Councilman bodies where the entire cell is shrunken into intense eosinophilic apoptotic body**

## PM1870. Causes of Leukocytosis

## PM1868. Immune Cell Antigens Detected by Monoclonal Antibodies

### PM1867. Anticoagulants in Vacutainers and their Uses

### PM1865. Ann Arbor Staging for Hodgkin and NonHodgkin Lymphoma

**All stages are further divided on the basis of the absence (A) or presence (B) of fever, night sweats and/or unexplained weight loss >10% of body weight.**

### PM1857. Pathophysiologic Classification of Polycythemia

**Pathophysiological classification of polycythemia**

### PM1792. Cancer Predisposition Syndromes and Associated Genes

### PM1740. Pathological Named Bodies

### PM1670. Differentiation of Mesothelioma from Adenocarcinoma

### PM1572. Defects in Leukocyte Function

## PM1571. Classification of Transplant Rejection Reactions

**Hyperacute rejection (minutes to hours):\n**

- **Mechanism: Antidonor antibodies present in the circulation of the recipient**
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### PM1568. Classification of Lupus Nephritis (International Society of Nephrology and Renal Pathology Society)

- **Focal- Involves only a few glomeruli**
- **Diffuse- Involves almost all the glomeruli**
- **Segmental- Involves only a part of glomerulus**
- **Global- Involves entire glomerulus**

### PM1566. Staining Patterns of Antinuclear Antibodies

### PM1562. Non–HLA Genes Associated with Autoimmune Diseases

### PM1553. HLA association with diseases

### PM1546. Innate and Adaptive Immunity

### PM1491. Classification of Iron Overload

### PM1480. Patterns of Drug- and Toxin-Induced Hepatic Injury

### PM1470. 2016 Criteria for Primary Myelofibrosis

**ET: Essential thrombocythemia\nPV: Polycythemia vera\nCML: Chronic myeloid leukemia\nLDH: Lactate dehydrogenase**

### PM1468. WHO Polycythemia Vera Criteria

**Major criteria:\n1. Hemoglobin >16.5 g/dL in men\nHemoglobin >16.0 g/dL in women\nOr\nHematocrit >49% in men\nHematocrit >48% in women\n2. Bone marrow biopsy showing hypercellularity with trilineage h**

### PM1467. Leucocyte Alkaline Phosphatase Score

**Alkaline phosphatase activity is found predominantly in mature segmented neutrophils. LAP score is obtained by assessing the staining intensity in 100 consecutive neutrophils, with each neutrophil sco**

### PM1466. Abnormalities Associated with Juvenile Myelomonocytic Leukemia (JMML) and Myelodysplastic Syndrome (MDS) in Children

**Juvenile myelomonocytic leukemia (JMML) is associated with all of the above.\nJMML:\n**

- **Formerly called juvenile chronic myelogenous leukemia**
- **Resembles Chronic myelomonocytic leukemia (**

### PM1465. 2016 Classification of MDS/MPN

### PM1464. 2016 Classification of myeloproliferative neoplasms

### PM1461. Classification of MDS

### PM1460. Bone Marrow Changes in Myelodysplastic Syndrome (MDS)

**The myelodysplastic syndrome can lead to dysplasia of the erythroid series, myeloid series, and megakaryocytes.\nDysplasia of erythroid series:\n**

- **Megaloblastoid maturation - A**

### PM1451. Blood Transfusion Reactions

**Non-immune mediated transfusion reactions are as follows:\n**

- **Transfusion-associated circulatory overload (TACO)**
- **Hypothermia**
- **Electrolyte abnorma**

### PM1404. Causes of Papillary Necrosis in the Kidney

### PM1399. EM findings in Renal Diseases

### PM1391. Glomerular Pathology in Renal Diseases

**Note:\n**

- **Anionic antigens form subendothelial deposits.**
- **Cationic antigens form subepithelial deposits.**
- **Neutral antigens form mesangial deposits.**

### PM1390. Immunoflouresence in Renal Lesions

**Image – Linear, smooth diffuse pattern of staining for IgG in Goodpasture syndrome.**

### PM1389. Types of Rapidly Progressive Glomerulonephritis

**Image: Crescentic glomerulonephritis (PAS stain) with the crescent-shaped mass of proliferating parietal epithelial cells and leukocytes internal to Bowman capsule.**

### PM1388. Complement Levels in Renal Diseases

**GN: Glomerulonephritis\nMPGN: Membranoproliferative glomerulonephritis\nSLE: Systemic lupus erythematosus\nHSP: Henoch–Schonlein purpura\nGBM: Glomerular basement membrane\nRPGN: Rapidly progressive g**

### PM1235. Cytogenetic Translocation and Associated Oncogenes Often Seen in Lymphoid Malignancies

### PM1207. Prognostic Factors in Acute Myeloid Leukemia (AML)

**TLC: Total leukocyte count\nMDS: Myelodysplastic syndrome**

### PM1193. Cytochemistry of Acute Leukemias

### PM1190. Immunophenotyping of Acute Lymphocytic Leukemia (ALL)

### PM1188. Panel of monoclonal antibodies for classification of leukemias

### PM1187. FAB classification

### PM1179. Morphological Characteristics of Blasts Cells in Acute Lymphoblastic Leukemia vs. Acute Myeloid Leukemia

**The image given below shows lymphoblasts. The image given below shows myeloblasts.**

### PM1178. Prognostic Factors in Acute Lymphoblastic Leukemia (ALL)

### PM1175. Predisposing Factors for Acute Lymphocytic Leukemia (ALL)

### PM1130. Polyposis Syndromes

### PM0956. Classification of Vascular Tumors and Tumor-like Conditions

### PM0953. Primary Forms of Vasculitis with Sites of Involvement

### PM0950. Classification of Vasculitis based on Pathogenesis

### PM0819. The Temporal Evolution of the Morphologic Changes in Acute MI

**EM: Electron microscopy\nLM: Light microscopy**

### PM0817. Approximate Time of onset of Key Events in Ischemic Cardiac Myocytes

### PM0815. Changes in the Aging Heart

**Small filiform processes form the closure lines of aortic and mitral valves, probably resulting from the organization of small thrombi - Lambl excrescences**

### PM0806. Named Cells in Histopathology

### PM0803. Gene Defects Associated with Congenital Heart Disease

### PM0694. Immunosecretory Disorders and Corresponding Neoplasms

### PM0692. Biochemical Forms of Amyloid and Associated Clinical Conditions

### PM0691. Prognostic Factors in Myeloma

### PM0690. WHO Classification of Plasma Cell Neoplasms

### PM0674. Lymphomas and Chromosomal Translocations

### PM0673. Immune Cell Antigens Detected by Monoclonal Antibodies

### PM0671. Origin of leukemia / lymphoma and their immunophenotypes

**CLL: Chronic lymphocytic leukemia\nSLL: Small lymphocytic lymphoma**

### PM0669. Lymphomas with Etiology

**EBV: Epstein–Barr virus\nHHV 8: Human herpesvirus-8\nHTLV: Human T-cell leukemia/lymphoma virus**

### PM0668. Lymphoid Neoplasms with their Cell of Origin

**CLP, Common lymphoid precursor; BLB, pre-B lymphoblast; DN, CD4/CD8 double-negative pro-T cell; DP, CD4/CD8 double-positive pre-T cell; GC, germinal-center B cell; MC, mantle B cell; MZ, marginal zone**

### PM0608. Pathological Cells/Bodies in Multiple Myeloma

- **Dutcher bodies - pale-staining, PAS-positive intranuclear inclusions within plasma cells containing Ig, which are more common in IgA myeloma**

### PM0538. Proteins of Bone Matrix

### PM0533. Skeletal Disease with Genetic Defects

### PM0473. Various Banding Techniques and stains used in Karyotyping

**The following are the banding techniques and stains used in cytogenetics to produce a visible karyotype by staining condensed chromosomes.**

### PM0410. Functions of Complement

- **Inflammation: C3a, C5a, and, to a lesser extent, C4a are called anaphylatoxins.**
- **C5a is also a chemotactic agent.**
- **Opsonization and phagocytosis:**

### PM0396. Epithelial Lesions and Risk of Malignancy

### PM0394. Factors Influencing Healing

- **Infection: one of the most important causes of delay in healing; prolongs inflammation; increases the local tissue injury**
- **Diabetes: most important s**

### PM0393. Acute phase reactants and their functions

### PM0389. Selectins and Associated Ligands

### PM0388. Integrins with Associated Ligands

### PM0385. Comparison of Acute and Chronic Inflammation

### PM0384. Diseases Caused by Inflammation and Cells Involved

**ARDS: Acute respiratory distress syndrome**

### PM0324. Causes of Disseminated Intravascular Coagulation (DIC)

**Disseminated intravascular coagulation (DIC) is an acute, subacute, or chronic thrombohemorrhagic disorder occurring as a secondary complication in a variety of diseases. It is characterized by the ac**

## PM0322. Embolism: Key Points

**Embolism: Key Points\n**

- **An embolus is a solid, liquid, or gaseous mass carried by blood to a site distant from its origin; most are dislodged thrombi.**
- **Pulmonar**

## PM0321. Thrombosis-Key Points

**Thrombosis-Key Points:**

**\n**

- **Thrombus development usually is related to one or more components of the Virchow triad:**
  - **Endothelial injury (e.g., by toxins, hyperte**

## PM0320. Arterial and Venous Thrombi

## PM0314. Hyperemia and Congestion

**Hyperemia and Congestion:**

## PM0313. Edema: Key Points

**Edema: Key Points\nEdema is the result of the movement of fluid from the vasculature into the interstitial spaces; the fluid may be protein-poor (transudate) or protein-rich (exudate).\nCauses:\n**

## PM0293. Enzymes as Tumor Markers

**Enzymes as tumor markers**

## PM0290. Protooncogenes, Oncogenes and Oncoproteins

**Protooncogenes, Oncogenes, and Oncoproteins\n**

- **Proto-oncogenes: normal cellular genes whose products promote cell proliferation**
- **Oncogenes: mutated or**

### PM0282. Inhibitors of Mitogenic Signaling Pathway

### PM0281. Cell Cycle Regulators and Associated Cancers

### PM0280. Cancers associated with important oncogenes

### PM0279. Protein involved in Signal transduction and Associated Cancers

**Protein involved in Signal transduction and Associated Cancers**

### PM0278. Growth Factor Receptors and Associated Tumors

**Growth factor receptors and Associated Tumors**

### PM0275. Etiological Agents for Cancers

### PM0273. Malignancies Associated With Paraneoplastic Polycythemia

- **Renal cell carcinoma**
- **Hepatoma**
- **Cerebellar hemangioma**
- **Adrenal cortical tumors**
- **Pheochromocytomas**
- **Virilizing ovarian tumors**
- **Wilms' tumor**

### PM0261. Malignancies, Translocations and Affected Genes

### PM0210. Important features of inheritance patterns

**X-linked Recessive Conditions:\n**

- **All sex-linked disorders are X-linked, and almost all are recessive.**
- **Several genes are located in the “male-specific region of Y;” al**

### PM0208. Ehler–Danlos Syndrome (EDS)

### PM0132. Apoptotic genes

### PM0130. Classification of Amyloidosis

### PM0129. Appearance of Amyloid

**H&E: Hematoxylin and eosin**

### PM0121. Important Stains in Pathology

### PM0120. Special stains used for staining Lipid

- **Oil Red O**
- **Sudan Black or Sudan IV**
- **Osmium tetroxide.**

**The image shows Oil Red O staining of lipid.**

## PM0119. Important Features of the Different Types of Hodgkin's Lymphoma (HL)

**Classic subtypes\n**

- **Nodular sclerosis**
- **Mixed cellularity**
- **Lymphocyte-rich**
- **Lymphocyte depletion**

**\nNon-Classical subtype\n**

- **Lymphocyte predominant**

## PM0117. Variants of Reed Sternberg Cells

**Variant Reed–Sternberg cells and the corresponding subtype of Hodgkin's Lymphoma\n**

- **Lacunar Variant: Nodular Sclerosis**
- **Mononuclear variant: Mixed cell**

## PM0116. Hodgkin v/s Non-Hodgkin Lymphoma

## PM0011. Types of Bone sequestra seen in chronic osteomyelitis

**Types of Bone sequestra seen in chronic osteomyelitis:**